

Appendix C: Cost Analysis

Appendix C.1 Summary Table

Component	Quantity	Total Cost
Platform	1	\$3,393.51
Safety Brace	1	\$712.77
Scissor Lift	1	\$1,474.39
Tower Clamping System	1	\$10,513.18
Total Cost		\$16,093.85

Appendix C.2 Individual Component Details

Appendix C.2.1 Platform

Raw Material	Purpose	Quantity	Cost	Catalogue
Aluminium round bars 10mm x 2meters	Aluminium round bars act as shafts for the guide wheels as well as the side safety arms.	1	\$10.57	[7]
RHS 100x50x2.0mm 12 m long	Provide the structural integrity for the platform. RHS are used to make the bottom and upper part of the platform strong and stable.	1	\$294.80	[6]
RHS 75x50x1.6mm 8 metres	These RHSs are used for the safety rails and arms.	1	\$185.00	[6]
Threadplate 1200x 2400x1.6mm Code: TP1.612003000 PP2TI	Threadplate is used on the base of the platform to help distribute the load of the person onto the RHSs as well as for aesthetics.	1	\$186.57	[13]
Sheet Metal 1200 x 1200 x 3mm Code:	Sheet metal is used for various parts. Such as brackets within the upper compartment of the platform, as well as for the side safety guide rails. Additionally,	2	\$572.86 (\$286.43 each)	[12]

C3.01200PV1FD	sheet metal is used for the back plates that cover up the motor.			
4140 Steel Round Bar 40 x 700mm Code: 4140RND.40	Cut and milled to create the two motor shafts	1	\$63.26	[18]
4140 Steel Round Bar 180 x 60mm Code: 4140RND.180	Cut and milled to create the two pinions	1	\$86.01	[18]
Bought Parts	Purpose	Quantity	Cost	Catalogue
75mm Diameter Guide wheels (note, these come with bearing already inside them) Code: CGZGRQ75BH	Guide wheels are used to keep the platform stable when it is attached to the racks. Some of the guide wheels, in an ideal situation, will not carry any load, unless in the event of severe weather forces. They take some loading and so durable guide wheels are chosen.	8	\$160.80 (\$20.10 each)	[11]
10X1.0 External Circlip DIN 471 Code: 10MEC	Circlips are used to keep bearings fixed and prevent them from coming off the shaft for the guide wheels and the side safety guides.	16	36.80 (\$2.30 each)	[8]
Compression Springs Code: C0420-055-2500	Compression springs are used for the guide wheels so that the guide wheels will stay in a reasonably fixed location. They were incorporated into the design as they can be adjusted (compressed) when first trying to engage the system with the racks. Note that these compression springs are quite long which is required.	8	\$98.32 (\$12.29 each)	[10]
Hinges Code: 72-7-3666	Hinges are used to open the door to the bottom storage compartment of the platform.	2	\$26.76 (\$13.38 each)	[14]
Latches	A latch is used to secure the bottom storage compartment	1	\$15.26	[14]

Code: 16-7-3681	when it is not needed to be opened. It allows for equipment and materials to be safely transported up and down the wind turbine in the event of good and poor weather conditions.			
SKF Deep Groove 10mm Bore Bearings Code: W 6000-2RS1	These bearings are used to allow for rotation in the side safety guide arms so that they can be raised and lowered.	4	\$29.08 (\$7.27 each)	[9]
General Purpose Buttons Code: 800Z-GF3Q5	These buttons will be used on the control bar to allow the engineer to transport up and down the wind turbine as well as switch it to remote control.	3	\$12.96 (\$4.32 each)	[16]
Booted Knobs Code: 800H-NPS	This will be used to turn the system on and off.	1	\$8.32	[16]
Emergency Stop Push-Pull button Code: 800T-FXLE6D4S	This button will allow the engineer to stop the scissor lift in the event of an emergency.	1	\$15.26	[16]
Shaft Pillow Block Bearing Assemblies Code: P 52 R-20 RM	Provide stabilisation of shaft, ensuring limited deflection. Assembly consists of: YET 204 (Bearing) P 52 (Housing) RIS 204 (Rubber seating ring)	2	\$117.78 (\$58.89 each assembly)	[17]
*Siemens SIMOGEAR CA29-LE80MH4 P Code: 2KJ3601-5DC23-9AQ1-Z	Power the shafts and pinions	2	\$906.50 (\$453.25 each)	[18]
20x1.2 External Circlips DIN471 Code: 20MEC	Circlips are used to ensure no axial movement of the pinion on the motor output shafts. These are for alignment and	4	\$12.80 (\$3.20 each)	[8]

	safety, as the interference fit between shaft and pinion should stop any slipping.			
Process	Purpose	Quantity	Cost	Catalogue
Welding	Welding is required to connect the base and upper RHSs to one another. It is also performed on the back safety guide as well as a means to secure guide wheel shafts to the RHSs. Majority of the platform is connected via welds and as such there are 28 individual welds in total required throughout the platform.	1	\$571.71	Calculated via solidworks
Cutting	A lot of the platform is made of RHS. As such, the majority of the cutting is simple end cuts (there are 38 end cuts). In addition to these cuts, some parts require more than end cuts, and instead use a mill to create slot cuts.	1	\$218.28	Calculated via solidworks
Plastic Injection Mould	The upper body of the platform is created via plastic injection moulding as it is light and will not be taking a large portion of the loading. Furthermore, it is done for aesthetic purposes as it makes the platform look more luxurious rather than industrial. Both the plastic injection mould operation and material cost of ABS will be outsourced and the total cost represents both the material and creation cost.	1	118.33	Calculated via solidworks
Drilling Holes	There are numerous holes throughout the base and the upper compartments of the platform.	1	\$23.06	Calculated via solidworks
Machining of Shaft	The shaft is constructed from a single block of material. Multiple different milling set ups and operations required to achieve required shape and tolerances of the shafts various geometries, such as the	2	\$346.12	Calculated via solidworks

	splines, circlip grooves, keyway and shaft stepping. The several setup costs contribute to the majority of the cost, while the processes themselves are relatively cheap. This causes the shaft machining price to dramatically decrease for mass production, however for this report only 2 shafts are being manufactured and thus setup costs dominate.			
Machining of Pinion	The pinions on each shaft are constructed from a single block of material. As with the shaft, multiple different milling operations are required to construct the shape. Cuts for the teeth, shaft bore and spline are required. Unlike the shaft the costs of conducting the milling processes are significant, therefore mass production will not reduce the cost to the same scale as the shaft	2	\$126.69	Calculated via solidworks
Fasteners	Connecting	Quantity	Cost	Catalogue
Rivets 6.4 X 25.0mm Code: ASB-12	These rivets are used for the upper part of the platform. They help to connect the plastic moulded upper body to the RHS. They also help fasten the brackets within the platform body that connect and secure the plastic moulded upper body to the RHS.	48	\$20.16 (\$0.42 each)	[15]
Rivets 4.0 X 8.4mm Code: AS5-03	These rivets are used to fasten the threadplate to the RHS on the base of the platform.	94	\$7.85 (\$0.09 each)	[15]
Phillips Pan Head Screws Gauge 8g Length 10	These screws are used to secure the back plates of the platform. Phillips are used as Phillips screwdrivers are most commonly available. We want this as we expect the back plates to be removed every now and then to perform check	28	\$18.05 (\$0.72 each)	[15]

	ups on the motors. These are also used to fasten the latch and the hinges to the platform.			
Hex Bolt M6X10	The metric bolts are used to hold down the motors and the bearings that connect to the shaft.	12	10.58 (\$0.88 each)	[15]

Total Cost	\$3,393.51
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* The motor cost does not need to be included in the final cost evaluation

Appendix C.2.2 Safety Brace

Raw Material	Purpose	Quantity	Cost	Catalogue
Alloy Steel Plate	Side panels, snap hook connector, push button and brake	1	\$66.15 (41.63 USD)	[20]
Bought Parts	Purpose	Quantity	Cost	Catalogue
520308 Snap Hook	Connect person to safety brace mechanism	1	\$4.72	[21]
91801A227 M4x0.7 Screws	Join the safety brace mechanism together	4	\$1.91	[22]
8969T856 15 mm compression spring	Ensure the push button stays closed	1	\$4.27	[23]
8 mm diameter steel cable	Allows connection of safety brace to wind tunnel	1	\$476.80	[24]
Process	Purpose	Quantity	Cost	Catalogue
Machining	Creating all of the parts for the safety brace that needs to be machined	1	\$158.92	Calculated via solidworks

Total Cost	\$712.77
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Appendix C.2.3 Scissor Lift

Bought Part	Purpose	Quantity	Cost	Catalogue
Scissor Lift Code:	Allows for the ease of transportation of the platform. Also allows for	1	\$1,474.39	[3]

HW1002	the platform to be easily attached to and disattached from the wall.			
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Total Cost	\$1,474.39
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Appendix C.2.4 Tower Clamping System

Bought Parts	Purpose	Cost (EA)	Quantity	Cost	Catalogue
Parallel flange beam, 12 m	Spacing between the system and the tower wall	\$503.85	20	\$10,077.07	[31]
* Vertical Rack	Integral to the pinion based climbing mechanism	\$802	158	\$126,716	[32]
Process	Purpose	Cost (EA)	Quantity	Cost	Catalogue
Hole drilling	Preparing Tower wall for fasteners	NA	240	\$14.95	SolidWorks calculation
Hole drilling	Preparing beam for fasteners	NA	714	\$24.13	SolidWorks calculation
Hole drilling	Preparing Racks for fasteners	NA	474	\$10.06	SolidWorks calculation
Fasteners	Connecting	Cost (EA)	Quantity	Cost	Catalogue
M10 x 45 Hex Bolt	Beam to tower wall	\$0.493	240	\$118.32	[25]
M6 x 70 Hex bolt	Vertical rack to beam	\$0.34	474	\$161.16	[26]
M10 Nyloc Nut	Beam to tower wall	\$0.212	240	\$50.88	[27]
M6 Nyloc Nut	Vertical rack to beam	\$0.051	474	\$24.174	[28]
M10 Washer	Beam to tower wall	\$0.034	480	\$16.32	[29]
M6 Washer	Vertical rack to beam, rack side only	\$0.034	474	\$16.116	[30]

Total Cost	\$10,513.18
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* The vertical rack cost does not need to be included in the final cost evaluation